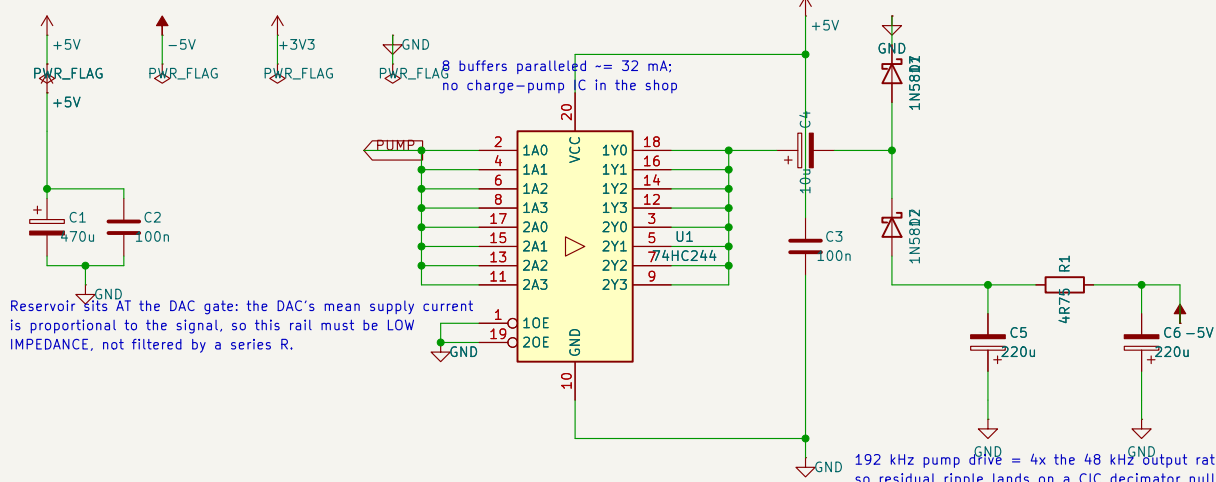


POWER (Pi +5V => charge=pump =5V => +/-2.5V DAC reference)



Reservoir V_{DD} at the DAC gate: the DAC's mean supply current is proportional to the signal, so this rail must be LOW IMPEDANCE, not filtered by a series R.

SPICE TESTBENCHES (sim/, one project each -- click to open)

A rails and the -5 V charge pump

B the +/-2.5 V ratiometric reference

C clock buffer and 74HC4040 divider

D integrator 1 with a real TL074

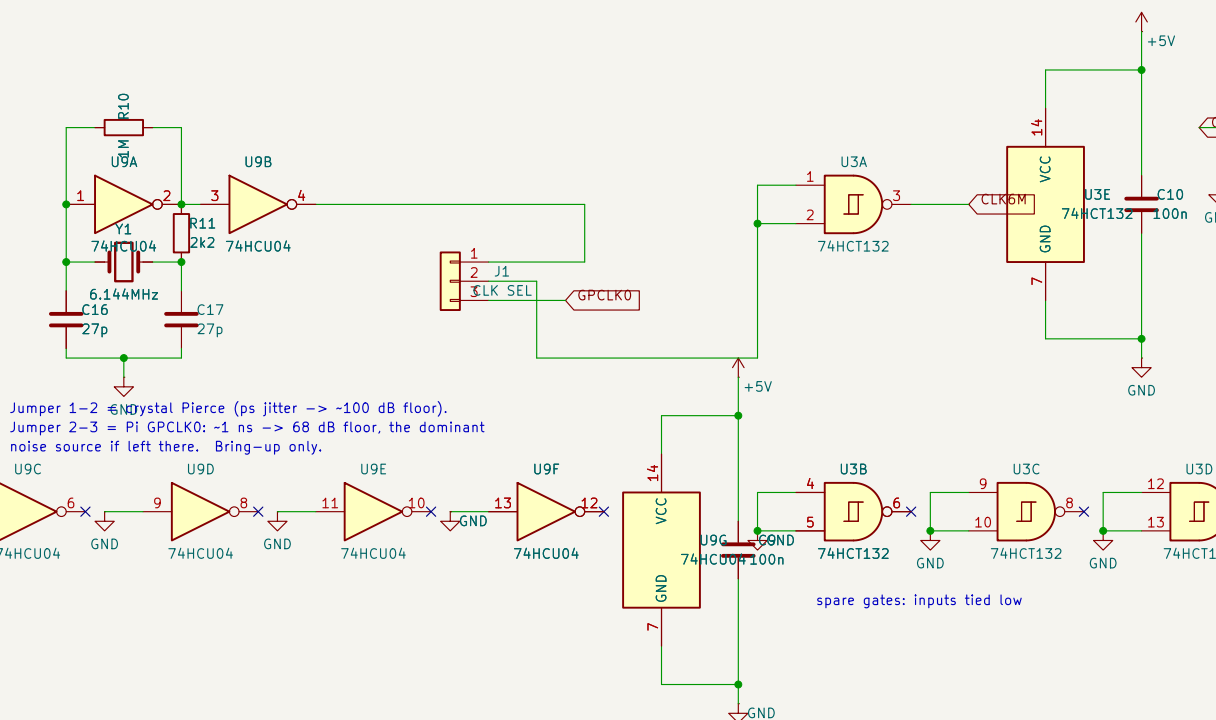
E LM311 quantiser on a single supply

F the 1-bit DAC and its offset leg

G retiming, interleave and the 3.3 V level shift

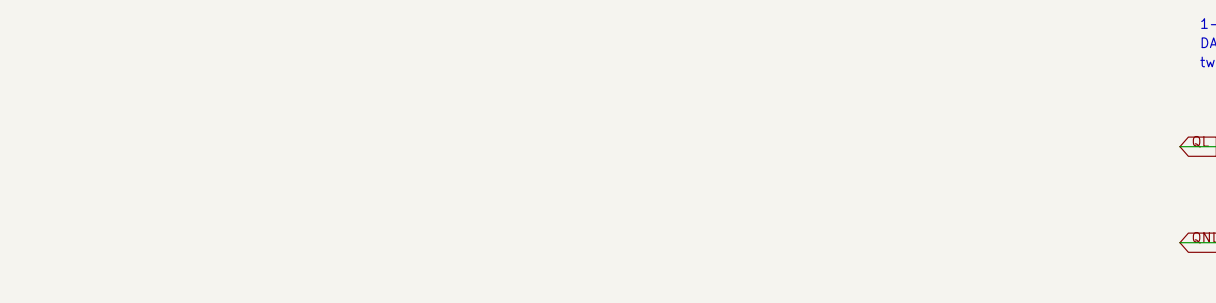
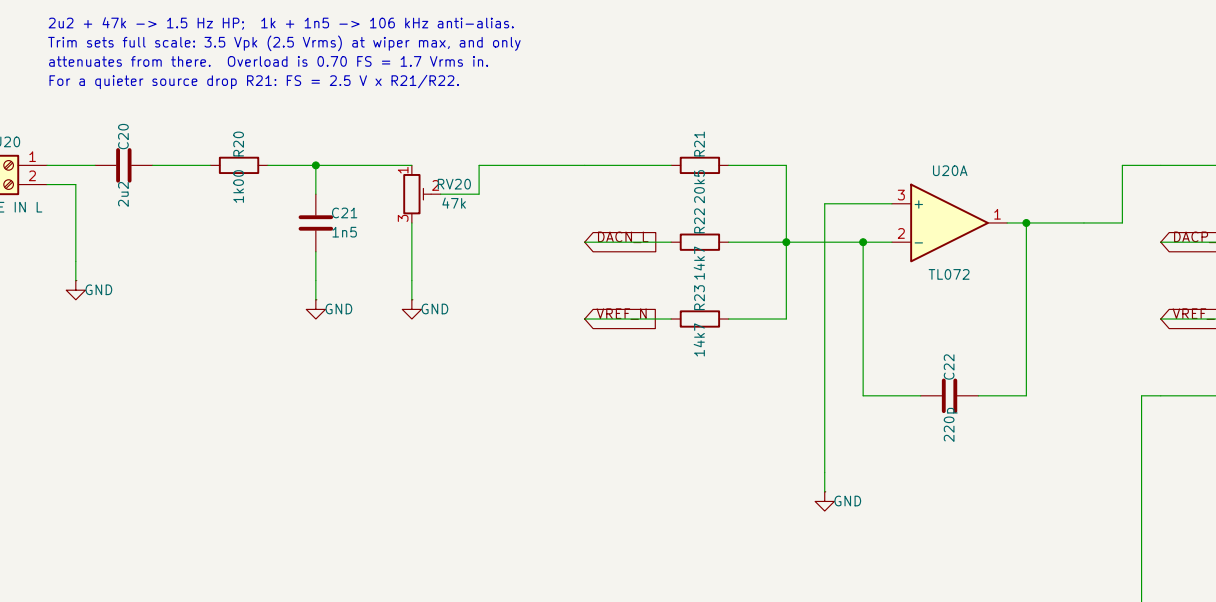
H one complete channel, with the click test

CLOCK AND DIGITAL (6.144 MHz => /2 BCLK 3.072M, /4 MCLK 1.536M, /128 LRCLK 48k; L and R interleaved onto one DIN)



CHANNEL L - 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32

INTEGRATOR 1 (a1 = 0.247)



CHANNEL R - 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32

INTEGRATOR 1 (a1 = 0.247)

